

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Engineered the backend document search engine in *Python (FastAPI)* for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Built backend *REST API* endpoints in *FastAPI* querying **600K+ documents** on *Elasticsearch* in **under 1 second**
- Designed and built a text-retrieval algorithm that extracts evidence segments from debate cases by validating source URLs and filtering statistical outliers with **IQR** on the word-count distribution of tag, citation and evidence text

Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)
- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson's CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

SchoolPulse AI | Live Website | Source Code

June 2026

- Built a *Python FastAPI* data pipeline that ingests live water, waste & energy data from 3 tools, runs **4 Random Forest models** for anomaly detection & delivers actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Citation Chain | Live Website | Source Code

June 2026

- Stored paper data in *PostgreSQL (Supabase)* and wrote *SQL* queries to filter papers by field, year and citation count
- Created a dual-API data pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Deployed a **RAG agent (Claude Sonnet 4.5)** that retrieves session-scoped vector documents (*Supernemory*)

Chronos | Live Website | Source Code

December 2025

- Designed a **5-step Python backend scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated it across **14 unit and integration test cases**
- Built a **scheduling agent (React/Firebase)** that auto-books meetings for large groups (30+ ppl) on Google Calendar
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

Truck Pricing VLM | Live Website | Source Code

September 2026

- Led a team of **4** to build a pricing algorithm for **Kamion (YC S22)**. **2nd of 7 at UIUC Founder's Hackathon**
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**
- Scraped **15,000** truck-listing images for *data mining*, auditing data quality via de-duplication and non-USD filtering

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, Java, C++, JavaScript, SQL, HTML/CSS, Next.js, React

Databases & Cloud: PostgreSQL, Elasticsearch, Docker, AWS, MySQL, Firebase, Supabase, Google Cloud, Vercel

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Grad-CAM, Hugging Face, Claude API, Gemma 4, RAG

Tools & Practices: API Development, Agile, VS Code, Claude Code, Codex, Figma, Kaggle, Google Calendar API

Concepts: REST APIs, Unit Testing, Data Pipelines, Full-Stack Development, Data Ingestion, ETL, Data Quality